

Vibe-IC — All Steps: Phase → Stage → Step

Every step of the full flow, organised as **Phase → Stage → Step** with a **single continuous numbering 1 → 44** (starting from Stage 1's Spec-to-RTL). Phase 1's document-generation steps are labelled **D1–D5** (pre-flow, not counted in 1→44); two parallel tracks — **Analog A1–A9** and **Mixed-signal M1–M4** — run alongside.

Phase → Stage map

- **Phase 1** — Specification & documents: two entries (**Agent path** • **doc-gen path D1–D5**) + optional architecture-exploration front-ends
- **Phase 2** — RTL → synthesis: Stage 1 (RTL+verification) • Stage 2 (constraints+synthesis+DFT+handoff gate)
- **Phase 3** — Physical → Tapeout: Stage 3 (physical+sign-off) • Stage 4 (output+tapeout) • Stage 5 (manufacturing & test)
- **Parallel** — Analog A1–A9 • Mixed-signal M1–M4

Phase 1 — Specification & documents

Two entries; both produce the same L-series design documents that feed Phase 2:

- phase1/input_prompt/ — free text / natural language → **Agent path**
- phase1/input_doc/ — existing documents / structured YAML → **doc-gen path** (D1–D5)

Agent path (free-text input)

Agent	What it does
IC Expert Agent	The single Phase-1 front door: faces the user directly in plain language, turns natural-language requirements into design facts (one plain-language question per gap), then reviews every layer with silicon expertise, fills sensible defaults, and runs cross-layer consistency checks.

Flow: user free text → IC Expert Agent → finalised L-series documents.

doc-gen path D1–D5 (existing-document input)

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
D1	Doc extraction → L1–L13	Ingest prompt/docs into input_doc/ and deterministically extract the core design layers.	user docs / prompt	L1 – L13 JSON	deterministic extractors	phase1_all_l_docs_present_check skills: phase1
D2	Core design-layer docs L1–L13	Deterministic extraction of datasheet, FRS, register map, etc.	D1 plain text	L1_DATASHEET ... L13	deterministic extractors	—
D3	Extended docs L14–L23	Protocol, timing, power intent (L21), skeletons.	L1–L13	L14 – L23 JSON	overlay extractor	—
D4	Protocol-class synthesis	Detect the IC's protocol class (81 classes) and synthesise class facts.	full input text	ic_class + protocol facts	is_ + _synth	—
D5	Coverage report	Verify the input documents landed completely in the L docs.	input docs + L docs	parity / coverage report	parity reporter	—

Architecture-exploration front-ends (optional, feed Step 1)

Front-end	What it does	Input	Output
architecture-explore	Design-space exploration (pipeline depth / parallelism / memory vs PPA, Pareto filter).	L docs + perf targets	architecture decisions (feed Step 1)
	C/C++/SystemC → RTL via HLS (open-source XLS, etc.).	C/SystemC model	RTL (verified from Step 2 onward)

Front-end	What it does	Input	Output
hls-c2rtl			
SpinalHDL/Chisel front-end	eda_spinalhdl_gen emits Verilog from Scala HDL.	SpinalHDL source	Verilog RTL

Front-end precedence (**artifact-driven**): existing RTL > C/SystemC model (hls-c2rtl) > SpinalHDL/Chisel > prompt-only (Step 1 spec-to-RTL). The available artifacts decide the path — never pick one speculatively.

Phase 2 — RTL → synthesis

Stage 1 — RTL generation & verification

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
1	Spec-to-RTL	Author synthesizable RTL from the L-series docs (SoC/CPU classes may take the IP-catalog reuse + glue path, e.g. Caravel-class harness platforms).	L1–L23 docs	rtl/*.v(.sv) • coverage report	— (AI-authored from L docs; SoC via IP-catalog)	skills: spec-to-rtl
2	 Lint (RTL hygiene)	Static RTL style/bug checks; auto-fixable issues fixed first.	RTL	lint reports (hygiene / ROM-init)	Verilator lint	rtl_hygiene_lint • rom_init_lint • rtl_bug_report_schema_check • internal_vs_external_timing_check... skills: rtl-review
3	 CDC / RDC check	Clock-domain / reset-domain crossing safety.	RTL	CDC/RDC reports (crossing / async / reset-dep)	in-house CDC/RDC scan	cdc_crossing_check • cdc_async_input_check • reset_dependency_check
4	 Simulation (testbench + coverage)	Per-IC oracle testbench functional simulation (golden compares) + coverage measurement; when L21 declares power domains, the TB should cover power-state transition scenarios (no open-source UPF-aware sim — structural verification stays with M2).	RTL • L10 test cases	sim logs • results.xml • coverage report	iverilog/vvp • Verilator coverage eda_simulate	testbench_gen • coverage_closure • l10_tb_conformance_check • l12_tb_coverage_check... skills: testbench-gen
5	 Formal verification (assertions)	Prove key properties hold: safety invariants proved unbounded and functional properties bounded-model-checked with the bound disclosed (all_l_proved requires the .sby + SymbiYosys evidence chain).	RTL • L3 constraints	.sby • formal results • full-stack TB results	SymbiYosys (ABC pdr / bmc3) eda_formal	formal_property_run • assertion_property_check • bit_level_full_stack_tb_check • formal_proof_evidence_check skills: formal-verify
6	FPGA early prototype	Pre-synthesis behavioral prototype on FPGA.	RTL • board constraints	.sof • map report • FPGA verification audit	Quartus eda_synth	fpga_test_harness_gen • debug_first_pass • quartus_map_audit • fpga_verification_audit

Stage 2 — Constraints, synthesis, DFT & handoff gate

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
7	Constraint setup (SDC + PVT matrix)	Author timing constraints (SDC) + PVT corner matrix; power intent modelled via L21 and rendered to UPF (handoff artifact — open-source tools do not	L8 timing • L21 • PDK liberty	*.sdc • pvt_matrix.json • <top>.upf (optional, when L21 declares power domains)	— (UPF emitted by the l21_to_upf_emit script, not an EDA engine)	sdcsyntax_check • pvt_matrix_check • l21_to_upf_emit • upf_syntax_check

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
		consume UPF; structural verification stays with M2).				
8	 SDC validation (incl. derived-clock guard)	Validate constraint correctness, completeness, and exception (false_path/multicycle) justification; manual ICG / register-divided clocks must declare a matching create_generated_clock (vacuous PASS without derived clocks — i.e. it passes automatically when the condition does not apply: no divided clocks in the design means the check self-skips).	SDC • L8 • RTL	SDC check report • derived_clock_sdc.json	—	sdcsyntaxcheck • sdcvalidatorcheck • sdcexceptioncorrelationcheck • derived_clock_sdc_requiredcheck
9	Synthesis (Yosys → mapped netlist)	Synthesize RTL and technology-map (dfflibmap + abc -liberty) to the standard-cell netlist.	RTL • SDC • liberty	synth/netlist.v • area stats	Yosys + abc eda_synth	synth_wrapper_gen • synth_netlist_check • provenance_check skills: synth-doctor
10	 Pre-layout STA (multi-corner)	Pre-layout static timing analysis (SS/TT/FF) + post-synth power preview (gate-level vectorless, default toggle rate, disclosed via analysis_mode; a different accuracy tier from Step 33's post-layout, optionally VCD-vector, analysis).	netlist • SDC • liberty	pre-PnR timing report + summary • pre_pnr_power_preview.rpt	OpenSTA eda_sta	sta_report_check skills: sta-review
11	DFT insertion (scan chain + ATPG)	Insert scan chains + generate patterns (open-source Fault: scan + stuck-at ATPG + TAP; MBIST/LBIST/compression out of open-source scope).	netlist	scan netlist • ATPG coverage report	Fault (scan+ATPG+TAP) eda_dft	fault_atpg_run • dft_atpg_coverage_check
12	Post-DFT optimization (resynth / buffering)	Re-optimize timing/area after DFT insertion.	scan netlist	post_dft_netlist.v	Yosys resynth	skills: synth-doctor
FS1	ISO-26262 FMEDA diagnostic-coverage (safety designs only)	Fault-injection on a declared safety mechanism (ECC/parity): inject stuck-at faults on the protected path and measure diagnostic coverage against the ASIL floor. Not applicable for non-safety designs.	RTL • declared safety mechanism • ASIL	fmEDA_coverage report (measured DC)	iverilog fault-injection	fmEDA_fault_injection_coverage • fmEDA_coverage_check
DT1	Transition-delay-fault (at-speed LOC) ATPG	Generate launch-capture 2-patterns for transition faults on a 2-time-frame launch-on-capture unroll of the scan-cut netlist, and grade the transition test coverage. Not applicable for combinational / no-scan designs.	scan-cut netlist • clock	transition coverage report	Yosys SAT (sat -prove)	transition_fault_atpg_run • transition_coverage_check
DT2	Path-delay-fault (at-speed, timing-graded) ATPG	Report the K longest launch-on-capture paths from the routed netlist with real post-layout timing, then generate and grade a launch-capture 2-pattern per path (robust and non-robust); a path with no possible pattern is excluded, never counted. Not applicable before the routed netlist and parasitics exist.	routed netlist • SPEF • SDC • scan cut	path-delay coverage report	OpenSTA + Yosys SAT (sat -prove)	path_delay_fault_atpg_run • path_delay_coverage_check
DT3	Small-delay-defect (SDD) at-speed grade	Grade each timing-critical path fault by its real post-layout slack: a defect is caught at-speed only when the detecting path's margin is tight, so detection through a low-slack path is a strong catch and through a slacky path is weak. A slack-rich design honestly scores low (its margin masks small delays). Descriptive grade, no floor.	path-delay coverage • transition coverage • SDC	SDD coverage report	OpenSTA + Yosys SAT (sat -prove)	sdd_atpg_run • sdd_coverage_check

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
		Not applicable before the path-delay and transition coverage exist.				
13	 Equivalence check (RTL $\equiv$ netlist)	Formally prove gate-level netlist $\equiv$ RTL (LEC).	RTL • post-DFT netlist	LEC report	Yosys equiv	lec_equivalence_check skills: equivalence-check
14	 Synthesis handoff gate (pre-PnR Yosys audit)	Synthesis→PnR handoff QA: synth script + netlist audit (open-source-Yosys specific ; the synthesis stage's closing gate).	synth script • netlist	handoff audit reports	Yosys script/netlist audit	yosys_hilomap_required_check • yosys_script_template_check

Phase 3 — Physical design → Tapeout

Stage 3 — Physical design & sign-off

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
15	Floorplan + PDN	Chip floorplan + power delivery network; tapcell insertion (latch-up well-ties, SKY130 14 μ m rule).	netlist • hardmacro LEFs (pdk_local/ auto-included, via the IP integration check: LEF/GDS/Liberty alignment + corner coverage + L21 supply consistency; macro LEFs should carry obstruction layers)	floorplan.def • PDN	OpenROAD (init_floorplan+pdngen+tapcell) eda_pnr	phase3_backend_step • floorplan_pdn_check • ip_integration_check
16	Clock planning	Clock-tree distribution strategy.	floorplan	clock_plan.json	OpenROAD CTS planning	clock_plan_check
17	Placement (global + detailed)	Place standard cells.	floorplan • netlist	placed.def	OpenROAD (global+detailed place) eda_pnr	placement_legality_check
18	Spare-cell + ECO-prep insertion	Pre-place spare cells / ECO readiness so later fixes are metal-only (paired with Step 32 ECO: provisioned here, consumed there).	placed.def	spare_cells.json • coverage report	OpenROAD eda_pnr	spare_cell_coverage_check • spare_cell_preservation_check
19	CTS (clock tree synthesis)	Build and balance the clock tree.	placed.def • clock plan	post_cts.def • clock-tree report	OpenROAD CTS eda_pnr	cts_quality_check
20	 Post-CTS hold fixing	Repair hold violations after CTS (the runner repeats hold repair post-global-route).	post_cts.def	post_hold.def	OpenROAD repair_timing -hold	hold_closure_check skills: hold-fix
21	Routing (global + detailed)	Complete all signal routing.	post_hold.def	routed.def • router DRC report	OpenROAD TritonRoute eda_pnr	drc_report_check • def_stage_progression_check • provenance_check
22	Parasitic extraction (SPEF)	Extract post-route parasitics; when the PDK ships no coupling captable, lateral coupling caps are added analytically from the routed geometry with a disclosed generic dielectric.	routed.def • tech LEF	SPEF (coupling-aware)	OpenRCX eda_extraction	spef_extraction_check • provenance_check
23	 Post-route STA (multi-corner sign-off)	Sign-off timing with real parasitics (MMMC = per-corner loop, one report per corner).	netlist • SPEF • SDC • multi-corner liberty	post-route timing report • per_corner/	OpenSTA (per-corner loop) eda_sta	sta_report_check skills: sta-review
24	 IR drop (static + dynamic)	Power-grid voltage-drop analysis vs budget: static, plus VCD-vectored dynamic IR (activity-weighted droop under a real switching VCD).	routed.def (PSM) • VCD	static + dynamic IR reports	OpenROAD PSM (read_vcd)	ir_drop_report_check • dynamic_ir_drop_check skills: ir-drop-triage
25	 EM check (electromigration)	Current-density / metal-lifetime screen.	routed.def (PSM -enable_em)	EM report + per-segment currents	OpenROAD PSM -enable_em	em_report_check
26	 Antenna check	Detect + repair process-antenna violations.	routed.def	antenna report	OpenROAD check_antennas/repair	antenna_report_check
27	 Signal integrity (crosstalk)		SPEF			si_crosstalk_check

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
		Crosstalk/noise screen (SPEF coupling-cap; advisory tier explicitly named).		SI report (incl. >0.9 coupling watch-list)	in-house SPEF coupling screen (OpenSTA window advisory)	
28	 PERC / Reliability sign-off (ESD + latch-up + cross-domain)	Enforced sign-off: ESD pad-ring + discharge topology, latch-up well-tap, cross-voltage-domain protection; maps to the four PERC categories — netlist checks + netlist-driven layout checks (automated), current density + P2P resistance (named manual-review). Manual review is signed off by a senior physical-design / reliability engineer; criteria live in <code>perc_equivalent.json</code> (<code>categories[].status=MANUAL_REVIEW</code> + <code>review_criteria</code> : PDK Jmax tables, foundry ESD discharge-path P2P limit, $V_{hold} > V_{dd}$, L21 cross-domain contract), results back-filled into <code>checklist[].confirmed</code> .	gate netlist • routed.def • L21 power intent • L3 ESD spec • step-24–27 reports	<code>perc_equivalent.json</code> • PERC memo • gate verdict	in-house PERC-equivalent (DEF-driven)	<code>perc_signoff_check</code>
29	Post-layout gate-level simulation (SDF)	Gate-level sim with SDF delays to confirm post-layout function (honest SKIP when no SDF re-sim ran).	gate netlist • SDF • TB	post-sim results	iverilog + SDF <code>eda_simulate</code>	<code>post_layout_sim_check</code>
30	Post-layout SPICE verification	Transistor-level ngspice correlation vs the Liberty timing: a representative cell plus the top-N STA critical paths, one worst path per distinct endpoint (extracted subckts stitched stage by stage with real net cap; SPICE vs STA path delay per path, aggregated).	SPICE decks • SPEF • STA paths	cell + top-N path SPICE correlation reports	ngspice + OpenSTA <code>eda_spice</code>	<code>spice_correlation_check</code> skills: <code>ams-sim</code>
31	 Physical verification (DRC + LVS + ERC + density)	Sign-off physical rules; density here is RULE COMPLIANCE (KLayout per-layer CMP-window deck; execution verification → Step 34, optimization advisory → Step 35); LVS = Magic extraction + real netgen compare (macro-bearing designs — e.g. Caravel-class harnesses — may blackbox macros: <code>Magic lef write -hide</code> masks the macro to its interface shell, a netgen supplementary setup compares it as a same-name blackbox; the waiver basis = device-level match + KLayout cross-check, written by <code>signoff_waiver_emit</code> into the project's <code>waivers.json</code> , cross-referenced as reviewer to-dos by the Step 36 checklist's open_waivers).	GDS • gate netlist • PDK decks	sign-off DRC • LVS • ERC reports	KLayout DRC • Magic ext2spice + netgen LVS • OpenROAD ERC <code>eda_drc_klayout</code> • <code>eda_lvs</code>	<code>erc_density_check</code> skills: <code>drc-fix</code> • <code>lvs-triage</code>
32	 ECO (repair loop)	Engineering-change repair loop when sign-off finds issues (consumes the Step 18 spare cells first, for metal-only fixes).	sign-off reports	ECO log or no-ECO flag	OpenROAD ECO	<code>eco_loop_audit</code> skills: <code>eco-plan</code>

Stage 4 — Output & Tapeout

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
33	Power analysis (post-layout)	Full-chip power sign-off (post-layout vectorless OpenSTA report_power; optional VCD vector mode).	netlist • SDC • liberty (+ optional VCD)	power report (leakage+dynamic, <code>analysis_mode</code>)	OpenSTA <code>report_power</code> (+ optional VCD)	<code>power_report_check</code>

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
34	Metal fill (density fill insertion)	Std-cell-row filler placement (white-space); density here is EXECUTION VERIFICATION (the metal_fill_density_check gate owns the density verdict; rule compliance → Step 31, optimization advisory → Step 35).	routed.def	filled.def • density report	OpenROAD filler_placement eda_pnr	metal_fill_density_check • spare_cell_preservation_check
35	DFM screen (manufacturability)	Manufacturability screen: redundant-via ratio (single-cut fraction advisory) + density OPTIMIZATION ADVISORY (cross-references the Step 34 gate result only — never a duplicate FAIL); OPC/RET/SRAF/PSM as FOUNDRY_SIDE disclosure items (escalated to DESIGNER_COLLAB_REVIEW at ≤28nm; the node is derived by dfm_screen_check from the input/pdk/Liberty filenames and recorded in the same dfm_screen.json — process_nm / advanced_node / foundry_side fields).	routed.def • density report	dfm_screen.json (via stats + foundry-side list)	in-house DEF via statistics + density cross-ref	dfm_screen_check
36	Tapeout checklist (final sign-off)	Item-by-item final confirmation (substance checks: DRC counts, evidence chains).	all sign-off reports	tapeout_checklist.json	— (inventory aggregation)	tapeout_signoff_check skills: tapeout-checklist
37	GDSII output	Stream the foundry-deliverable GDSII (only when Step 31 PV is fully clean).	routed.def • merged GDS	sign-off *.gds	Magic/KLayout stream-out eda_gds	gds_size_check • provenance_check
38	Foundry handoff (mask spec + WAT + scribe + corner vectors)	Foundry physical mask kit: mask spec + WAT plan + scribe PCM + corner ATE vectors (chip-specific; foundry-supplied fields named PENDING_FOUNDRY_* — tracked in the Step 36 checklist and back-filled after the foundry replies).	GDS • netlist stats • L10 cases	mask_spec.json • wat_plan.json • scribe • corner_test_vectors.json	— (pack generator)	foundry_handoff_package_check skills: tapeout-checklist
39	FPGA final sign-off (on-board test)	Final FPGA recompile + on-board attestation with hardware evidence.	RTL • board	final .sof • on_board_pass.json	Quartus + on-board measurement	bringup_plan_gen • fpga_on_board_attestation_check

Stage 5 — Manufacturing & test (post-fab; triggers on silicon receipt)

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
40	Fabrication (foundry, external)	Foundry mask-set + wafer fab (OPC/RET are foundry-side mask synthesis).	foundry handoff kit	mask/wafer intake attestations	external foundry	manufacturing_fab_intake_check
41	Wafer sort / probe test	Wafer probing, good-die selection; yield independently re-derived vs target.	wafer lot • probe card	wafer_sort_yield.json • wafer_map.csv	ATE + probe card (external)	wafer_sort_yield_check
42	Packaging (assembly)	wirebond / FC-CSP / WLCSP assembly.	good dies	packaging_log.json	assembly house (external)	packaging_intake_check
43	Final test (ATE + burn-in)	Post-package final test (functional + parametric + burn-in infant-mortality screen).	packaged units • ATE patterns	final_test_yield.json • burn_in_results.json	ATE (external)	final_test_attestation_check
44	Reliability qualification (HTOL / FIT)	Long-duration HTOL qual (device-hours / failures / FIT attestation; required for automotive/medical grades; consumer MPW may stay dormant = DEFERRED, never blocks tapeout).	HTOL chamber results	htol_results.json verdict	HTOL chamber (external)	htol_attestation_check

Out-of-flow lab steps (unnumbered): PFA/EFA (destructive FIB/SEM/EMMI failure analysis) and silicon characterization (shmoo) — data originates from external equipment; the plugin owns the data-analytic root-cause layer (wafer_map_pattern_classify, yield-diagnostic).

Parallel tracks

Analog A1–A9 (parallel to Stages 1–3)

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
A1	Analog spec extraction	Extract per-block analog specs from the L docs.	L5 ADI spec	spec.json	deterministic spec extract	analog_a1_spec_extract_check skills: analog-spec-extract
A2	Topology selection	Choose the circuit topology per spec.	spec.json	topology.md	AI topology select	analog_a2_topology_select_check skills: analog-topology-select
A3	Netlist generation	Generate the SPICE netlist.	topology • PDK models	<block>.sp	xschem netlist eda_xschem_netlist	analog_netlist_pdk_check skills: analog-netlist-gen
A4	Corner sweep (PVT)	PVT corner sweep + Monte-Carlo yield (mc_yield_pct ≥ 95% gate).	.sp • PDK corner libs	corner_results.json (executed/ derived counts)	ngspice (corner + MC) eda_spice_corner	analog_a4_corner_sweep_check skills: analog-sizing-loop
A5	Analog layout	Complete the analog layout.	netlist	layout.mag / GDS	Magic layout eda_analog_layout	analog_a5_layout_check skills: analog-layout
A6	Block physical verification (per-block DRC + LVS)	Per-block DRC + LVS before merge (catches block-level errors top-level PV would mask).	block GDS • netlist	DRC clean / LVS match flags	Magic DRC + netgen LVS	analog_a6_block_pv_check skills: drc-fix • lvs-triage
A7	 Post-layout resimulation	Re-simulate with extracted parasitics; compare pre-vs-post specs (>10% degradation loops to A3).	layout parasitics • spec	pre_vs_post.json	Magic extraction + ngspice eda_spice_corner	analog_pre_vs_post_layout_check skills: analog-extraction-resim
A8	Hardmacro generation (LEF + Liberty + GDS + Verilog)	Package the block for Stage-3 consumption (LEF should carry obstruction layers — Magic lef write -hide or abstract with obs — so top-level routing cannot enter the macro).	layout • characterisation	4-file hardmacro	Magic/abstract + characterisation	analog_hardmacro_check skills: analog-hardmacro-gen
A9	 Co-simulation / HW verification	Mixed digital+analog simulation and hardware-in-the-loop verification.	hardmacro • digital RTL	cosim / HW measurement results	iverilog + ngspice co-sim eda_spice • eda_simulate	mixed_signal_cosim_check • analog_hw_spice_correlation_check skills: mixed-signal-cosim • analog-hw-tuning-loop

Mixed-signal M1–M4 (triggered when analog blocks exist)

Trigger timing: runs once **A8 hardmacros are complete** and **Stage 3 is near closure** (routed/GDS mergeable) — hardmacros must exist before floorplan, but M1's GDS merge waits for the digital side to finish routing. If a hardmacro slips until Stage 3 has entered Step 31, M1 waits for the final hardmacro GDS and Step 31's top-level LVS re-runs (incremental — scoped to the macro-interface change).

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
M1	Top-level integration (A+D GDS merge)	Merge digital + analog GDS, place macros; run top-level LVS on the merged GDS (Magic extraction + netgen, macros blackboxed).	digital GDS • hardmacro GDS	top_merged.gds • merge/LVS report	KLayout merge + Magic/netgen top-LVS	mixed_signal_merge_check skills: analog-flow-orchestrate
M2	Power domain verification (level shifter / isolation)	Structural verification of cross-domain level-shifters / isolation, plus a cross-power-domain signal-crossing check that derives the isolation/level-shifter requirement from the power-domain definitions and audits the UPF strategy.	L21 • merged design	power_domain / level_shifter / isolation / signal-crossing reports	structural check programs	power_domain_crossing_check • level_shifter_required_check • isolation_cell_required_check • power_domain_signal_crossing_check skills: ir-drop-triage
M3	Mixed-signal verification (AMS co-sim)	AMS co-simulation + interface signal integrity.	merged design • cosim TB	cosim results • interface SI report	AMS co-sim	

#	Step	What it does	Input	Output	Tools (EDA)	Programs / Skills
						<code>mixed_signal_cosim_check</code> • <code>mixed_signal_interface_si_check</code> skills: <code>mixed-signal-cosim</code> • <code>ams-sim</code>
M4	Mixed-signal sign-off	Top-level verdict rolled up from M1–M3.	M1–M3 reports	<code>signoff.json</code>	rollup verdict	<code>mixed_signal_signoff_check</code> skills: <code>tapeout-checklist</code>

Totals

Phase	Stages	Steps
Phase 1 — Specific ation & docum ents	two entries (Agent • doc-gen) + architecture front-ends	D1–D5 + IC Expert Agent
Phase 2 — RTL → synthes is	Stage 1 • Stage 2	1–14
Phase 3 — Physica l → Tapeou t	Stage 3 • Stage 4 • Stage 5	15–44
Parallel	Analog • Mixed-signal	A1–A9 • M1–M4

44 sequential steps (Stage 1: 1–6 • Stage 2: 7–14 • Stage 3: 15–32 • Stage 4: 33–39 • Stage 5: 40–44), plus Phase 1 (Agent path & doc-gen path D1–D5) and the two parallel tracks (Analog A1–A9 • Mixed-signal M1–M4). Preflight: P0 (environment health check). Conditional lettered steps: FS1 (ISO-26262 FMEDA diagnostic-coverage, safety designs only) • DT1 (transition-delay-fault ATPG, scan designs only) • DT2 (path-delay-fault at-speed ATPG, scan designs after routing) • DT3 (small-delay-defect at-speed grade, after DT2). Orchestrator `vibe_ic_one_shot_runner.py` runs Phase 1 → Phase 2 → Analog → Phase 3.

Out of scope (declined with reasons): designer-EXECUTED OPC/RET (mask synthesis is foundry-side — surfaced as FOUNDRY_SIDE items in Step 35 + noted at Step 40), commercial hardware emulators (FPGA path covers the intent), MBIST/LBIST/EDT compression (no open-source engines), BSR/BSDL, automatic clock gating (no characterized ICG cell in sky130; manual RTL clock gating remains available — the SDC declaration of divided/gated clocks is guarded by Step 8's `derived_clock_sdc_required_check`), via-doubling/CAA (commercial DFM).

E1–E3 external-lab steps (reserved numbering, outside the 44; activated on automotive/medical grade upgrades):

#	Step	Notes
E1	PFA / EFA	FIB / SEM / EMMI failure analysis (external lab)
E2	Silicon characterization	shmoo plots • voltage/frequency sweeps
E3	Temperature cycling / mechanical stress	JESD22-class environmental stress (external lab)

Appendix: measured wall-clock reference (for intuition)

Numbers come from REAL local sky130 open-source-flow runs (the `duration_s` fields of `reports/orchestrator/*_one_shot.json`; design sizes 0.5k–21k cells) — measured, never estimated. Durations vary with design, constraints and machine; nonlinear in cell count.

Stage	Measured range	Samples
Step 9 Synthes is (Yosys)	~0.4 s – 21 s	0.5k cells (lpc) → 20k cells (cv32e40p)

Stage	Measured range	Samples
Steps 15–22 PnR (OpenR OAD, full)	~3 s – 31 min	3.4k cells (subservient) → 21k cells (sha256)
Step 37 GDS write	~2 – 4 s	all samples
Step 31 DRC (KLayout sky130 deck)	~8 – 84 s	0.5k → 20k cells
Step 31 LVS (Magic + netgen)	this batch's samples took the light/skip path, so no representative figure; with real macro compares (e.g. Caravel-class harnesses) it runs minutes to hours , scaling with macro count (a field run motivated the 4-hour timeout)	—
Steps 40–43 manufaturing (fab/sort/pkg/final test)	external, weeks-scale	no local measurement

Licensing & IP: the whole flow relies on open-source tools only (Apache-2.0 project; commercial-tool firewall and output ownership: repo-root README.md § IP ownership, plus the DCO/patent pledge in CONTRIBUTING.md).

繁體中文版：ALL_STEPS_v1.4.14.zh-TW.md.